

PORTABLE OPERATION · SITE, SIGNAL, DEBRIEF

FIELD RECORD

The Portable Operator's Deployment Log



SAMPLE "NOCALL" STATION

26 DEPLOYMENTS · FIELD REFERENCE · STATION RECORDS

FIELD RECORD

A logbook built around the outing, not the contact.

SAMPLE "NOCALL" STATION



FIELD RECORD

The Portable Operator's Deployment Log

26 DEPLOYMENTS · SITE, SIGNAL, AND DEBRIEF
FIELD REFERENCE FOR PORTABLE OPERATION

SAMPLE "NOCALL" STATION



FIELD RECORD: The Portable Operator's Deployment Log

Compiled and designed by Sample "NOCALL" Station.

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Frequency allocations, repeater listings, tone codes and program rules are facts and belong to nobody. The selection, arrangement, drawings and wording here are original to this work. Sources are named on the page that uses them and in the Colophon.

BUILD

This copy was generated on 08 September 2026. Regulatory figures were current when the sources in the Colophon were compiled. A copy printed long after that date should be checked against the rules before it is trusted.

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PART ONE

FIELD REFERENCE

The material a portable operator reaches for with cold hands. Read it once at the kitchen table. After that, use the markers in the outer margin.

KX2 QUICK REFERENCE

KX3 QUICK REFERENCE

IC-705 QUICK REFERENCE

IC-706MKII QUICK REFERENCE

IC-7000 QUICK REFERENCE

IC-7100 QUICK REFERENCE

IC-7300 QUICK REFERENCE

FT-818ND QUICK REFERENCE

FT-857D QUICK REFERENCE

FT-891 QUICK REFERENCE

FT-897D QUICK REFERENCE

FT-991A QUICK REFERENCE

G90 QUICK REFERENCE

X5105 QUICK REFERENCE

X6100 QUICK REFERENCE

KX2

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT AN ELECRAFT PUBLICATION AND NOT AFFILIATED WITH ELECRAFT.

MENU AND PANEL KEYS

WHAT THE BUTTONS DO

RATE + A/B	Held together about two seconds, this is the power switch. There is no separate one.
RATE / FREQ	Tap steps the tuning rate. Hold for direct frequency entry.
MODE / RCL	Tap changes mode. Hold enters memory recall, longer starts a scan.
BAND / STORE	Tap changes band. Hold stores a memory.
A/B / A>B	Tap swaps the VFOs. Hold copies A into B.
RIT / SPLIT	Tap toggles receive incremental tuning. Hold toggles split.
DISP / MENU	Tap cycles the alternate displays. Hold opens the menu.
XMIT / TUNE	Tap keys the transmitter. Hold sends a carrier for tuning.
ATU / PFn	Tap runs or selects the tuner. Hold reaches the programmable function.
PRE / NR	Tap toggles the preamp. Hold turns on noise reduction.
FIL / APF	Tap enters filter adjust. Hold gives the audio peaking filter on CW, or auto notch on SSB and AM.
DATA / TEXT	Tap selects data mode and sub-mode. Hold opens the decode display.
MSG / REC	Tap plays a stored message. Hold records one.
AF / MON	Receive audio level. The alternate function is monitor level.
KYR / MIC	Keyer speed on CW, mic gain on voice. Hold it to reach the power setting.

MENU ITEMS WORTH KNOWING

HOLD DISP TO ENTER

AGC MD	Turns AGC off so RF gain can be set by hand. The icon changes to AGC minus.
AGC SPD	Slow or fast. Fast is the default on CW, slow elsewhere.
RF GAIN	Manual receiver gain, for use with AGC off.
TX CMP	Speech compression on SSB.
VOX GN	Set so VOX triggers on normal speech, not on the dog.
CW WGHT	Keyer element to space ratio.
CW IAMB	Iambic A or iambic B.
MSG RPT	Auto repeat interval, 1 to 255 seconds.

FILTERS

PASSBAND, VARIABLE

Tap FIL to enter filter adjust. AF sets width, KYR shifts the passband, tapping AF normalises to the standard setting for the mode.

APF

Hold FIL on CW for a very narrow audio peak. Weak signal work only.

AUTO NOTCH

Hold FIL on SSB or AM to pull down a steady carrier.

NB

Noise blanker, for repetitive noise: power line, appliance, ignition. Use the lowest setting that works.

NR

Noise reduction, for hiss. It sounds hollow, and high settings eat weak signals.

SPECIFICATIONS

COVERAGE, RECEIVE	500 kHz to 32 MHz, general coverage
COVERAGE, TRANSMIT	80 through 10 m, nine bands
MODES	USB, LSB, CW and data, with PSK31, PSK63 and RTTY built in
OUTPUT	10 W on 80 through 17 m, 8 W on 12 and 10 m, per the product page
SUPPLY	9 to 15 V at the DC jack. Full SSB output wants 12 V or better
CURRENT, TRANSMIT	1 to 2 A typical
CURRENT, RECEIVE	About 150 mA with the backlight and preamp off
ANTENNA CONNECTOR	BNC, 50 ohms
WEIGHT	About 13 oz bare, without battery or options
INTERNAL TUNER	Optional KXAT2, rated 15 W
COMPUTER CONTROL	ACC jack, 3.5 mm, serial through a KXUSB or KXSER cable

COMMON OPERATIONS

STORE A MEMORY

Hold STORE, rotate VFO A to the channel number, hold STORE again to commit. Any other key abandons it.

RECALL A MEMORY

Hold RCL, pick the memory with VFO A, tap any key to leave.

SCAN

Put the start and end frequencies in VFO A and B, store that as a memory, recall it, then hold RCL about three seconds until SCAN appears. It stops on a modulated signal.

SET SPLIT

Hold A>B twice so B matches A, tune B to the offset, then hold SPLIT. The TX icon moves to VFO B.

SET OUTPUT POWER

Hold the KYR / MIC knob to bring up PWR, then rotate.

KX2

FIELD REFERENCE, CONTINUED

OPERATING NOTES

- Thirteen ounces is the whole argument for this radio. If the rest of the kit is not chosen to match, the weight saving goes back into the pack somewhere else.
- Receive current in the low hundreds of milliamps means a small battery lasts a long day, as long as you are not transmitting flat out for all of it.
- The internal tuner is an option. Know whether yours has one before you plan an antenna that depends on it.
- Elecraft's manual and product page disagree on maximum output. Set the power you actually use, measure it once, and write it below.
- Log the output you used. On a radio with this range, five watts and ten watts are different activations.

COMPUTER CONTROL

JACK

ACC, 3.5 mm stereo, on the left side

CABLE

KXUSB for a computer USB port, KXSER for a real RS-232 port. A bare cable will not do.

BAUD

Set with the BR command. The choices are 4800, 9600, 19200 and 38400. Write yours below.

ALSO ON THIS JACK

Firmware updates and the KXPA100 amplifier.

NOTES ON THIS RADIO

STATION SETTINGS

FILL THIS IN ONCE

SERIAL BAUD RATE

MEASURED OUTPUT, SSB

MEASURED OUTPUT, CW

INTERNAL TUNER FITTED

FILTERS AND PASSBAND

CW PITCH AND SIDETONE

KEYER SPEED AND WEIGHTING

MIC GAIN

BREAK-IN SETTING

MEMORY BANK SCHEME

RESET PROCEDURE, FROM THE MANUAL

FIRMWARE VERSION

MEMORY PLAN

WRITE IT DOWN BEFORE YOU NEED IT

[illegible]

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KX3

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT AN ELECRAFT PUBLICATION AND NOT AFFILIATED WITH ELECRAFT.

MENU AND PANEL KEYS	WHAT THE BUTTONS DO
BAND- + TUNE	Held together about two seconds, this is the power switch.
BAND+ / BAND-	Steps band up or down.
MODE / ALT	Tap picks the mode. Hold picks the alternate: CW reverse, or the other sideband.
DATA	Tap enters data mode. Tap again to step the sub-modes.
FREQ ENT	Numeric frequency entry.
A/B / REV	Tap swaps the VFOs. Hold reverses them temporarily.
RATE / KHZ	Tap picks 1 or 10 Hz. Hold gives the coarse rate. A long hold locks VFO A.
SPLIT	Hold to enable split transmit.
RIT / XIT, CLR	Tap enables the offset. Hold CLR to zero it.
PBT I/II	Tap selects low cut or high cut on voice, width or shift on CW and data. Hold normalises.
PRE / ATTN	Preamp, or the 15 dB attenuator.
NB / NR	Noise blanker and noise reduction.
APF / NTCH	Audio peak on CW, auto notch on SSB and AM. Hold NTCH for the manual notch.
KEYER / MIC	Keyer speed in words per minute on CW, mic gain on voice. Hold it to reach the power setting.
PITCH / DLY / CMP	Sidetone pitch, VOX or break-in delay, and compression. Holding PITCH on FM turns on the repeater tone.
XMIT / TUNE	Key the transmitter, send a tuning carrier, or start a tuner cycle.
MENU	Hold to enter. The offset knob scrolls entries, VFO A edits the value.

MENU ITEMS WORTH KNOWING	HOLD MENU TO ENTER
RPT OFS	FM repeater offset.
VOX GN	VOX trigger level.
CW WGHT	Keyer element to space ratio.
CW IAMB	Iambic A or iambic B.
MIC BTN	What the microphone buttons do.
TX GATE	Mutes mic audio below a threshold.
BAT MIN	Voltage at which the low battery warning appears. Worth setting for the pack you actually use.
BKLIGHT	Backlight on or off. It costs current.

FILTERS	
PASSBAND, VARIABLE	PBT I/II sets low and high cut on voice, width and shift on CW and data. Hold to normalise.
KXFL3 ROOFING FILTER	Optional analog module, 3000 Hz and 1000 Hz, selected automatically from the mode and filter settings.
APF	Audio peaking filter for weak CW.
NB AND NR	Blanker for impulse noise, reduction for hiss.
SPECIFICATIONS	
COVERAGE, RECEIVE	1.5 to 32 MHz general coverage, 0.31 to 1.5 MHz reduced, plus 6 m
COVERAGE, TRANSMIT	160 through 6 m. 2 m or 4 m with the optional module
MODES	SSB, CW, data with four sub-modes, AM, FM
OUTPUT	10 W on most bands, 8 W on 12, 10 and 6 m. Current firmware raises this to 15 W on 80 through 20 m above 12.8 V supply
SUPPLY	9 to 15 V DC. 12 to 14 V for full output
CURRENT, TRANSMIT	1 to 3 A, varying with band, supply and load
CURRENT, RECEIVE	0.15 to 0.25 A
ANTENNA CONNECTOR	BNC for 160 through 6 m. The VHF module has its own SMA
WEIGHT	About 18 oz without options
INTERNAL TUNER	Optional KXAT3, wide range, 20 W
COMPUTER CONTROL	ACC1 jack, 3.5 mm, serial through a KXUSB or KXSER cable

COMMON OPERATIONS	
STORE A MEMORY	Hold STORE, then rotate VFO A to the memory you want.
RECALL A MEMORY	Hold RCL, pick the channel with VFO A, tap any key to exit.
SCAN	Put the start and end frequencies in VFO A and B, then hold SCAN.
SET SPLIT	Hold SPLIT.
SET OUTPUT POWER	Hold the KEYER / MIC knob to reach PWR, then rotate.

KX3

FIELD REFERENCE, CONTINUED

OPERATING NOTES

- Fifteen watts out of a set of AA cells is a short conversation. Plan the power source around the mode and the length of the outing, not around the maximum the radio will make.
- The AA socket is a real field advantage because cells can be bought anywhere. Log which chemistry you used: alkaline and nickel metal hydride behave differently under a keyed load.
- Six meters is on this radio. Worth a listen on a summer afternoon before you pack up.
- The base manual dates from 2014 and the errata does not. If a setting on your radio does not match this page, the errata is the document to read.
- The internal tuner is an option. Know whether yours has one.

COMPUTER CONTROL

JACK

ACC1, 3.5 mm stereo

CABLE

KXUSB to a computer USB port, KXSER to a real RS-232 port

BAUD

Set with the BR command. The choices are 4800, 9600, 19200 and 38400. Write yours below.

SECOND JACK

ACC2, 2.5 mm, is a keyline and logic output. It is not a control port.

UTILITY

The KX3 Utility is needed for configuration and firmware, and gives a CW and data terminal.

NOTES ON THIS RADIO

STATION SETTINGS

FILL THIS IN ONCE

SERIAL BAUD RATE

FIRMWARE VERSION

ERRATA REVISION READ

MEASURED OUTPUT, SSB

MEASURED OUTPUT, CW

INTERNAL TUNER FITTED

ROOFING FILTER FITTED

CW PITCH AND SIDETONE

KEYER SPEED AND WEIGHTING

MIC GAIN

BREAK-IN SETTING

BATTERY TYPE IN USE

LOW BATTERY WARNING SET TO

MEMORY BANK SCHEME

MEMORY PLAN

WRITE IT DOWN BEFORE YOU NEED IT

[illegible]

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IC-705

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT AN ICOM PUBLICATION AND NOT AFFILIATED WITH ICOM.

MENU AND PANEL KEYS

WHAT THE BUTTONS DO

POWER	Short press on, hold off.
MENU	Opens the settings tree, where the set items live.
FUNCTION	The per-contact surface: AGC, split and the like. Separate from MENU.
QUICK	Context menu for whatever is on the screen. Faster than MENU for most things.
EXIT	Leaves a screen or steps back one level.
MULTI	Push to open the multi-function menu, rotate to change the selected item.
VOX / BK-IN	Cycles VOX, or on CW cycles semi break-in, full break-in and off. Hold about a second to set the delay.
RIT / dTX	Tap turns the offset on and off. Hold swaps between receive and transmit offset.
XFC	Held, listens on the transmit frequency in split, or opens the squelch on simplex.
SCAN	Tap opens scan select. Hold about a second restarts the last scan.
MPAD	Tap steps the memo pad. Hold saves what is displayed into it.
CALL / DR	Tap swaps call channel and VFO. Hold turns the D-STAR repeater function on or off.
SPEECH / LOCK	Tap announces the frequency aloud. Hold locks the dial.
PBT	Push to swap PBT1 and PBT2, rotate to shift the passband, hold to clear.
AF / RF-SQL	Rotate for audio. Push to bring up its setting menu.

WHERE THE FIELD SETTINGS LIVE

HOLD PTT AND PUSH MULTI

RF POWER

Hold the PTT and push MULTI, then touch RF POWER. The scale is 0 to 100 per cent of whatever the supply allows.

MIC GAIN

Same multi-function menu, holding the PTT.

KEY SPEED

Multi-function menu. Roughly 6 to 48 words per minute.

CW PITCH

Multi-function menu. Roughly 300 to 900 Hz.

AGC

On the FUNCTION screen. Touch to step fast, mid and slow. Touch and hold to set the constant per mode.

BW

On the FILTER screen. Touch BW then rotate.

MAX TX PWR

FUNCTION screen. Caps transmit power, which is worth setting for a small battery.

FILTERS

FIL1, FIL2, FIL3

Three filter settings per mode. Touch the filter icon to step them, or hold it a second to open the filter screen.

BW

On the filter screen, touch BW and rotate to set the passband width of the selected filter.

PBT1 AND PBT2

Twin passband tuning. Push PBT to swap between them, rotate to shift, hold to clear.

SPECIFICATIONS

COVERAGE, RECEIVE	0.030 to 199.999 MHz and 400 to 470 MHz
COVERAGE, TRANSMIT	HF amateur bands, plus 6 m, 2 m and 70 cm
MODES	USB, LSB, CW, RTTY, AM, FM, DV. WFM receive only
OUTPUT, EXTERNAL DC	0.1 to 10 W, 0.025 to 2.5 W AM
OUTPUT, INTERNAL PACK	0.1 to 5 W, 0.025 to 1.25 W AM. The supply sets the ceiling, not a menu
SUPPLY	13.8 V DC external, or 7.4 V from the BP-272 pack
CURRENT, TRANSMIT	Under 3 A at 13.8 V, under 2.5 A at 7.4 V
CURRENT, RECEIVE	0.5 A at 13.8 V at full audio, 0.3 A in standby
ANTENNA CONNECTOR	One BNC, 50 ohms, all bands
WEIGHT	About 2.4 lb with the battery
INTERNAL TUNER	None. The external AH-705 is driven from the tuner jack
COMPUTER CONTROL	Micro USB, plus built-in Bluetooth and wireless LAN

IC-706MKII

OPERATING REFERENCE

INDEPENDENT OPERATING REFERENCE. NOT AN ICOM PUBLICATION AND NOT AFFILIATED WITH ICOM.

MENU AND PANEL KEYS

DISPLAY CYCLES THE SET

M1	SPL	Split. Long press to enable; transmit moves to VFO B.
M1	A/B	Swap VFO A and VFO B.
M1	MW	Memory write. Hold two seconds to store the current VFO.
M2	V/M	VFO and memory toggle. M-CH dial selects the channel.
M2	M>V	Copy the selected memory into the VFO. Memory clear in memory mode.
M3	FIL	Filter select. Cycles the filters actually installed.
M3	NB	Noise blanker. Built for ignition noise, not for hash.
M4	AGC	AGC fast and slow. SSB and AM only.
M4	RIT	Receive incremental tuning. Clear it before you forget it is on.
S1	TS	Tuning step.
S2	SCN	Scan start and stop. Set the squelch first.
S3	BAND	Band stack. Hold a band key to recall its stored frequencies.
G1	SET	Set mode. Long press DISPLAY to enter from any mode.

QUICK SET

LONG PRESS DISPLAY

Q1	RF POWER	Set it per band and per site. Write it in the Site Record.
Q2	MIC GAIN, CW PITCH, RTTY TONE	Changes meaning with the mode.
Q3	SQUELCH AND RF GAIN	Squelch closed before scanning.
Q6	REPEATER TONE	The encode tone the repeater wants.
Q7	tone SQUELCH	Only opens on a matching tone. Different from Q6.

FILTERS

FIL CYCLES WHAT IS FITTED

SSB WIDE	The default. Everything you work on phone.
SSB NARROW, OPTIONAL	Worth fitting. Turns a crowded 40 m evening into a workable one.
CW NARROW, OPTIONAL	Worth fitting if you work CW at all. It is the difference between hearing one signal and three.
FM	Fixed. Repeater work only.
AM	Wide. Shortwave listening, not much else.

SPECIFICATIONS

COVERAGE, RECEIVE	0.030 to 199.999 MHz, general coverage
COVERAGE, TRANSMIT	160 through 6 m, plus 2 m. No 70 cm on the MKII.
OUTPUT, HF AND 6 M	5 to 100 W, adjustable
OUTPUT, 2 M	5 to 20 W, adjustable
MODES	SSB, CW, AM, FM
SUPPLY	13.8 V DC, 20 A at full HF output
ANTENNA CONNECTOR	SO-239, single, all bands
REMOTE	CI-V on the REMOTE jack, 3.5 mm

COMMON OPERATIONS

STORE A MEMORY
VFO mode. Set frequency, mode, offset and tone. Pick a channel with the M-CH dial. Hold MW for two seconds.

WORK A REPEATER
Set the output frequency, the shift, and the encode tone in Q6. Listen before you call. On a linked system wait a beat after keying.

SPLIT FOR DX
Set your transmit frequency in VFO B, long press SPL. Check where you are actually transmitting before you call.

RESET A STUCK RADIO
Partial reset keeps memories. Full reset does not. Do the partial one first, and only reach for the full reset when you have written your memories down.

SET THE CLOCK
Set mode. UTC, always. Every field in this book assumes it.

FAULT FINDING

IN THIS ORDER

NO TRANSMIT ON A BAND YOU SHOULD HAVE
Check the band edge first, then the mode, then whether the tuner is faulted. The radio will refuse quietly.

REPEATER WILL NOT OPEN
Encode tone in Q6, not tone squelch in Q7. They are different settings and Q7 will make you deaf as well as silent.

COMPUTER WILL NOT SEE THE RADIO
CI-V address, then baud, then transceive, then the cable. In that order, one at a time.

RECEIVE IS DEAF ON ONE BAND
Check the preamp and the attenuator before you suspect the antenna.

DISPLAY SHOWS NONSENSE
Partial reset. Write your memories down first, every time.

IC-706MKII

OPERATING REFERENCE, CONTINUED

OPERATING NOTES

- Run it at the power the contact needs, not the power the radio has. Every watt above what works is amp-hours you do not get back.
- The finals get hot on a long CW or digital run with no airflow. Give it a fan or give it a rest.
- It draws about 20 A key down at 100 W. Fuse and wire for that even if you never use it.
- The internal tuner is an option, not a fitting. Know which one you have before you plan an antenna around it.
- Detachable head means the connector is a failure point. Carry the spare separation cable or accept that a bad day ends early.

CI-V AND COMPUTER CONTROL

JACK

REMOTE, 3.5 mm on the rear panel

ADDRESS

Read it from set mode on the radio. The 706 family uses a different address per model, and the MKIIG is 58h.

BAUD

Set mode. 19200 is the usual choice; drop it if the link is flaky.

TRANSCERVE

Set mode. On means the radio announces changes unasked.

LEVEL CONVERTER

CI-V is not RS-232. Use a CT-17 or an equivalent interface; a bare serial cable will not work.

NOTES ON THIS RADIO

STATION SETTINGS

FILL THIS IN ONCE

CI-V ADDRESS

CI-V BAUD

TRANSCIVE ON OR OFF

FILTERS INSTALLED

CW PITCH

MIC GAIN

SSB POWER, FIELD

CW POWER, FIELD

BREAK-IN SETTING

TUNER FITTED

SEPARATION CABLE

MEMORY BANK SCHEME

PARTIAL RESET KEY SEQUENCE

MODEL VARIANT

MEMORY PLAN

WRITE IT DOWN BEFORE YOU NEED IT

[illegible]

Written from the radio and its manual by K4DIA. Menu behavior differs on the MKIIG. Icom, IC-706 and IC-706MKII are trademarks of Icom Inc. This card is an independent operator's crib sheet, written from the radio and its manual, and is not published by, endorsed by or affiliated with Icom. Menu behavior varies between the MKII and the MKIIG. Check yours.

IC-7000

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT AN ICOM PUBLICATION AND NOT AFFILIATED WITH ICOM.

MENU AND PANEL KEYS

WHAT THE BUTTONS DO

MENU / GRP	Short push cycles the menus within a group, M-1 to M-3. Hold about a second to change group.
F-1 to F-4	Soft keys. What they do depends on the menu showing above them.
FUNCTION, QUICK, EXIT	None of these keys exist on this radio. Everything goes through MENU and the soft keys.
DIAL, TS, MODE	Main dial, tuning step, and mode.
PBT / M-CH / RIT	One concentric control doing all three.
AF and RF / SQL	Concentric pair.
BAND up / down	Steps the band.
TUNER / CALL	External tuner, or the call channel.
SPCH / LOCK	Speech, and panel lock.
ANF, MNF, NB, NR, P, AMP / A	Auto notch and recorder, manual notch, blanker, noise reduction, preamp and attenuator.

SET MODE GROUPS

HOLD MENU TO CHANGE GROUP

Quick set RF POWER, MIC GAIN, SSB TRANSMIT BANDWIDTH

Also keyer speed, CW pitch, sidetone level, the twin peak filter, and the RTTY settings.

Miscellaneous QUICK SPLIT, SPLIT OFFSET AND LOCK

Also the repeater offset per band, one touch repeater, auto repeater, tuner settings, and the CI-V address, baud and transceive.

Display set TWENTY-ONE ITEMS

Contrast through to the power-on check.

FILTERS

SSB TRANSMIT BANDWIDTH

Wide, mid or narrow, with separate low and high edges, in quick set mode.

TWIN PEAK FILTER

For RTTY.

SPECIFICATIONS

COVERAGE, RECEIVE	0.030 to 199.999 MHz and 400 to 470 MHz
COVERAGE, TRANSMIT	160 through 10 m, plus 6 m, 2 m and 70 cm
MODES	USB, LSB, CW, RTTY, AM, FM, WFM receive only. No D-STAR
OUTPUT	2 to 100 W on HF and 6 m, 2 to 50 W on 2 m, 2 to 35 W on 70 cm
SUPPLY	13.8 V DC plus or minus 15 per cent
CURRENT, TRANSMIT	About 22 A at full power
CURRENT, RECEIVE	1.3 A standby, 1.6 A at full audio
ANTENNA CONNECTORS	Two SO-239, 50 ohms
WEIGHT	About 5.1 lb
INTERNAL TUNER	None. An external AH-4 or AT-180 is the option
COMPUTER CONTROL	Remote jack only. There is no USB port on this radio

COMMON OPERATIONS

STORE A MEMORY

Hold the memory write key about a second to write the displayed frequency and mode into the shown channel.

SET SPLIT

The split soft key in the first menu row. Hold it about a second for quick split, once the offset is set in miscellaneous set mode.

SET OUTPUT POWER

RF power in quick set mode.

WORK A REPEATER

In FM, the shift soft key in the third menu row selects the direction, or off. Hold it for one touch repeater. The tone soft key selects encoder, tone squelch or digital squelch; hold to set the frequency or code.

OPERATING NOTES

- No USB port. A level converter is not an accessory on this radio, it is the only way a computer talks to it.
- No internal tuner, and two antenna sockets. The same two things to check as the IC-7100.
- Twenty-two amps at full output from a radio small enough to take anywhere is the trap. Fuse and wire for the current, not the size.
- The detachable head is a connector, and connectors fail. Carry the spare cable.
- This radio is out of production. Spares and heads come from the used market, which is a reason to write your settings down below.

COMPUTER CONTROL

CI-V JACK

The remote jack on the rear. There is no USB port

CI-V ADDRESS

70h by default

BAUD

300 up to 19200, or auto. Auto is the default

TRANSCIVE

On by default

LEVEL CONVERTER

A CT-17 or equivalent is required. With no USB port there is no way around it

IC-7000

FIELD REFERENCE, CONTINUED

STATION SETTINGS

FILL THIS IN ONCE

CI-V ADDRESS, AS READ FROM SET MODE

CI-V BAUD

ANTENNA SOCKET, PER BAND

SSB TRANSMIT BANDWIDTH

CW PITCH

KEYER SPEED

MIC GAIN

SSB POWER, FIELD

CW POWER, FIELD

REPEATER OFFSET, PER BAND

EXTERNAL TUNER

RESET PROCEDURE, FROM THE MANUAL

FIRMWARE VERSION

NOTES ON THIS RADIO

MEMORY PLAN

WRITE IT DOWN BEFORE YOU NEED IT

CH	FREQUENCY	MODE	SHIFT / TONE	WHAT IT IS
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Compiled from the Icom IC-7000 Instruction Manual and published specifications, September 2026. Not checked against a radio. The reset procedures could not be read and are left out. Icom and the model names here are trademarks of Icom Inc. This card is an independent reference and is not published by, endorsed by or affiliated with Icom.

IC-7100

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT AN ICOM PUBLICATION AND NOT AFFILIATED WITH ICOM.

MENU AND PANEL KEYS		WHAT THE BUTTONS DO
PWR / AF		Push on, hold about a second off. Rotate for volume.
MENU		Cycles the touch key sets, M-1 to M-3. It does not open a settings tree.
SET		Enters and leaves configuration.
QUICK		Opens the quick menu. Push again to close.
EXIT		There is no EXIT key on this radio. Back out with MENU or the return symbol on screen.
M-CH / BANK		Selects memories, or drives passband and offset when those are active.
MIC / RF PWR		One shared display for mic gain and transmit power.
SPEED / PITCH		Keyer speed, roughly 6 to 48 words per minute, and CW pitch, roughly 300 to 900 Hz.
SPEECH		Push announces the frequency and meter. Hold about a second for lock.
XFC		Listens on the transmit frequency.
TUNER / CALL		External tuner, or the call channel.
NB, NR, NOTCH, P.AMP ATT		The receiver conditioning keys, as printed.

TOUCH KEY ROWS		MENU CYCLES THEM
M-1	SCAN, SPLIT, A/B, V/M, MW	The row you will use most: split, VFO swap, memory write.
M-2	DUP, AGC, TONE, DSQL, VOICE, KEYS, DEC, COMP, RTTY, CS, TBW	Repeater shift and tone live here, along with AGC and the keyer.
M-3	MEMO, SCOPE, SWR, DTMF, VOX, BK-IN	Break-in and the standing wave meter.
D-1, D-2	D-STAR	The repeater rows in DR mode.

SPECIFICATIONS	
COVERAGE, RECEIVE	0.030 to 199.999 MHz and 400 to 470 MHz
COVERAGE, TRANSMIT	HF, 6 m, 2 m and 70 cm. US models have no 4 m band
MODES	USB, LSB, CW, RTTY, AM, FM, DV. WFM receive only
OUTPUT	2 to 100 W on HF and 6 m, 2 to 50 W on 2 m, 2 to 35 W on 70 cm
SUPPLY	13.8 V DC plus or minus 15 per cent
CURRENT, TRANSMIT	22 A on HF and 6 m, 16 A on 2 m and 70 cm
CURRENT, RECEIVE	1.2 A at full audio, 0.9 A standby
ANTENNA CONNECTORS	Two SO-239. One for HF and 6 m, one for 2 m and 70 cm
WEIGHT	Main unit about 2.3 kg, controller about 0.5 kg
INTERNAL TUNER	None. An external AH-4, AT-180 or AH-740 is the option
COMPUTER CONTROL	USB port, and a remote jack

COMMON OPERATIONS	
STORE A MEMORY	The memory write key in the first touch row.
RECALL A MEMORY	V/M in the first touch row. The channel knob steps channels.
SCAN	The scan key in the first touch row.
SET SPLIT	The split key in the first touch row.
SET OUTPUT POWER	The mic and power key, then adjust.
WORK A REPEATER	In FM, the shift and tone keys in the second touch row.

OPERATING NOTES	
	- No internal tuner, despite the size of it. Whatever you put up has to be resonant or matched by something you carried in. - Two antenna sockets, and the wrong one is the first thing to check when a band seems dead. - There is no EXIT key. If you are used to an IC-7300 you will hunt for it. - Twenty-two amps on HF. Size the supply for that even if you never run it. - The detachable head is a connector, and connectors fail. Carry the spare cable.

COMPUTER CONTROL	
CI-V JACK	The remote jack on the rear, or over USB
CI-V ADDRESS	88h by default
BAUD	Selectable from 300 up to 19200, or auto. Write yours below
LEVEL CONVERTER	A CT-17 is needed for the remote jack. USB does CI-V with no converter

IC-7100

FIELD REFERENCE, CONTINUED

STATION SETTINGS

FILL THIS IN ONCE

CI-V ADDRESS, AS READ FROM SET MODE

CI-V BAUD

ANTENNA SOCKET, PER BAND

FILTER WIDTHS

CW PITCH

KEYER SPEED

MIC GAIN

SSB POWER, FIELD

CW POWER, FIELD

BREAK-IN SETTING

REPEATER SHIFT AND TONE

EXTERNAL TUNER

RESET PROCEDURE, FROM THE MANUAL

FIRMWARE VERSION

NOTES ON THIS RADIO

MEMORY PLAN

WRITE IT DOWN BEFORE YOU NEED IT

CH	FREQUENCY	MODE	SHIFT / TONE	WHAT IT IS
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Compiled from the Icom IC-7100 Instruction Manual and published specifications, September 2026. Not checked against a radio. Figures adjusted to the US model. The reset procedures could not be read and are left out. Icom and the model names here are trademarks of Icom Inc. This card is an independent reference and is not published by, endorsed by or affiliated with Icom.

IC-7300

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT AN ICOM PUBLICATION AND NOT AFFILIATED WITH ICOM.

MENU AND PANEL KEYS

WHAT THE BUTTONS DO

MENU	Opens the settings tree: set, keyer, decode and the rest.
FUNCTION	Operating parameters for the current mode. A separate surface from MENU.
QUICK	A short list of the things most often changed on whatever is showing. Faster than MENU.
EXIT	Closes the screen and steps back a level.
MULTI	Rotate to set the assigned function, push to pick items.
A/B	Swaps the VFOs. Hold to equalise them.
V/M	Switches VFO and memory mode.
MPAD	Cycles the memo pad. Hold to store what is displayed.
NB, NR	Tap toggles. Hold about a second to set the level.
NOTCH	Cycles auto, manual and off.
P.AMP / ATT	Cycles the preamp. Hold about a second for the attenuator.
SPLIT	Tap toggles. Hold about a second for quick split.
TUNER	Engages the internal tuner.
XFC	Held, listens on the transmit frequency while split.
TWIN PBT CLR	Hold about a second to reset the passband.
Touchscreen	A brief touch beeps once. About a second gives the selection beep.

SPECIFICATIONS

COVERAGE, RECEIVE	0.030 to 74.8 MHz
COVERAGE, TRANSMIT	160 through 10 m, plus 6 m. US models have no 4 m band
MODES	SSB, CW, RTTY, AM, FM
OUTPUT	2 to 100 W. AM 1 to 25 W
SUPPLY	13.8 V DC plus or minus 15 per cent
CURRENT, TRANSMIT	21 A at 100 W
CURRENT, RECEIVE	0.9 A standby, 1.25 A at full audio
ANTENNA CONNECTOR	One SO-239, 50 ohms
WEIGHT	About 9.3 lb
INTERNAL TUNER	Yes, built in and automatic
COMPUTER CONTROL	USB Type-B, and a 3.5 mm remote jack. Both carry CI-V

COMMON OPERATIONS

SELECT MEMORY MODE

Press V/M. The display shows memory.

STEP MEMORIES

The up and down arrow keys.

MEMO PAD

Hold MPAD to capture the displayed frequency.

SET SPLIT

Set VFO A, press A/B, set VFO B, press SPLIT. Or hold SPLIT about a second for quick split.

SET OUTPUT POWER

Push MULTI, touch RF power, rotate. The scale is 0 to 100 per cent.

RESET, MEMORIES KEPT

Partial reset clears the transceiver settings and leaves your memory channels alone.

RESET, MEMORIES ERASED

All reset takes everything back to factory, memory channels included. Photograph them first.

OPERATING NOTES

- Twenty-one amps at full output. Size the battery, the fuse and the wire for that, then run the power the contact needs.
- It has a real internal tuner, which is the main thing separating it from the IC-7000 and IC-7100 in the field.
- The waterfall is the reason people buy this radio and the reason they sit watching instead of calling.
- Menu paths have moved between firmware revisions. If a path on this card is not on your radio, your firmware is the reason.
- One antenna socket, HF and 6 m only. There is no VHF or UHF on this radio.

COMPUTER CONTROL

CI-V JACK

The 3.5 mm remote jack on the rear, or over USB

CI-V ADDRESS

94h by default

BAUD

Auto by default, following the connected controller

TRANSCIVE

On by default, so the radio announces changes unasked

LEVEL CONVERTER

Not needed over USB. A CT-17 is only for the remote jack path

IC-7300

FIELD REFERENCE, CONTINUED

STATION SETTINGS

FILL THIS IN ONCE

CI-V ADDRESS, AS READ FROM SET MODE

CI-V BAUD

USB SERIAL FUNCTION

FILTER WIDTHS, FIL1 FIL2 FIL3

CW PITCH

KEYER SPEED

MIC GAIN

SSB POWER, FIELD

CW POWER, FIELD

BREAK-IN SETTING

MEMORY BANK SCHEME

FIRMWARE VERSION

NOTES ON THIS RADIO

MEMORY PLAN

WRITE IT DOWN BEFORE YOU NEED IT

CH	FREQUENCY	MODE	SHIFT / TONE	WHAT IT IS
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FT-818ND

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT A YAESU PUBLICATION AND NOT AFFILIATED WITH YAESU.

MENU AND PANEL KEYS

WHAT THE BUTTONS DO

PWR	Press and hold one second to switch on or off.
F	Tap shows the current function row. Hold one second enters the menu.
A, B, C	Act on whatever function row is showing. There are eleven rows, scrolled with SEL after tapping F.
SEL	Detented. Tunes, selects memories, and picks the function row.
MAIN dial	Tuning, and setting menu values.
V/M	Switches between the VFO and the memory system.
BAND up / down	One band per tap.
MODE up / down	Steps the mode.
HOME	Recalls the home memory.
CLAR	Tap turns on the clarifier. Hold half a second turns on IF shift.
LOCK	Locks the front panel keys.
SQL / RF	RF gain or squelch, whichever a menu item has assigned it to.
TX / BUSY lamp	Green when the squelch opens, red on transmit.

FUNCTION ROWS

TAP F, THEN SEL TO SCROLL

Row 1 A/B, A=B, SPL	VFO swap, equalise, and split.
Row 2 MW, MC, TAG	Memory write, memory check and skip, and channel naming.
Row 4 RPT, REV, TON	Repeater shift, reverse, and the CTCSS encoder.
Row 9 PWR, MTR	Transmit power, and what the meter shows.
Row 10 VOX, BK, KYR	VOX, break-in, and the internal keyer.

FILTERS

YF-122S

Optional Collins SSB filter, 2.3 kHz.

YF-122C

Optional Collins CW filter, 500 Hz.

YF-122CN

Optional Collins CW filter, 300 Hz. The one that changes a crowded evening.

SPECIFICATIONS

COVERAGE, RECEIVE	100 kHz to 30 MHz, 50 to 54 MHz, 76 to 154 MHz, 420 to 470 MHz
COVERAGE, TRANSMIT	160 through 6 m, plus 2 m and 70 cm, amateur bands only
MODES	SSB, CW, AM, FM and digital
OUTPUT	6 W from external 13.8 V. Four steps: 6, 3.5, 2.5 and 1 W
OUTPUT, ON BATTERIES	The radio forces 2.5 W on the internal pack or eight AA cells
SUPPLY	8 to 16 V DC, and the supply must hold 3 A
ANTENNA CONNECTORS	Two. Rear SO-239 and front BNC. Which one is live per band is a menu setting
RECEIVER	Double conversion, first IF 68.33 MHz, second 455 kHz. 1 W of audio into 8 ohms
INTERNAL TUNER	None
COMPUTER CONTROL	ACC jack, 8-pin mini DIN, shared with band data and cloning

COMMON OPERATIONS

STORE A MEMORY

Tap F, SEL to the memory row, tap MW to enter memory check, SEL to the channel, then hold MW one second until the double beep.

STORE A SPLIT MEMORY

Store the receive frequency as above. Set the transmit frequency in the VFO, tap MW, and while the channel number blinks, close the PTT while holding MW one second.

RECALL A MEMORY

Tap V/M once. SEL picks the channel. V/M again returns to the VFO.

SCAN

Tap SCN to start scanning upward. Tap again to stop.

SET SPLIT

Tap F, SEL to the VFO row, A/B to select VFO B, tune it, then tap SPL.

SET OUTPUT POWER

Tap F, SEL to the power row, tap PWR to step through the four levels.

WORK A REPEATER

Tap F, SEL to the repeater row. RPT selects minus then plus shift; hold RPT one second to reach the shift size. TON turns on the encoder; hold TON one second to reach the tone frequency.

OPERATING NOTES

- Six watts is the whole radio. Everything else about operating it follows from that: the antenna has to be better than the one you would get away with at a hundred watts.
- The supply must hold 3 A even though the radio only makes six watts. A supply that sags will not just cut power, it will make the audio sound wrong on transmit.
- Two antenna jacks, and a menu item decides which one is live per band. It is the first thing to check when a band appears to be dead.
- On internal cells the radio drops to 2.5 W and does not ask. If your log shows six watts and you were on batteries, the log is wrong.

FT-818ND

FIELD REFERENCE, CONTINUED

CAT AND COMPUTER CONTROL

RATE

Menu. The radio and the program must agree. If the program connects but reads nonsense, the rate is wrong before anything else is.

HANDSHAKE

Menu. Leave it off unless the program asks for it.

INTERFACE

Older sets need a level converter between the CAT jack and a computer. Later ones present a USB port and enumerate as two serial ports, one for control and one for audio.

PTT

Decide once whether the computer keys the radio over CAT or over a separate line, and write it down. Half the digital problems in the field are two things trying to key at once.

STATION SETTINGS

FILL THIS IN ONCE

CAT BAUD RATE

ANTENNA JACK, PER BAND

FILTERS FITTED

REPEATER SHIFT SIZE

CTCSS TONE

MIC GAIN

KEYER SPEED

SSB POWER, FIELD

CW POWER, FIELD

MEMORY BANK SCHEME

RESET PROCEDURE, FROM THE MANUAL

FIRMWARE VERSION

NOTES ON THIS RADIO

MEMORY PLAN

WRITE IT DOWN BEFORE YOU NEED IT

Compiled from the Yaesu FT-818ND Operating Manual, September 2026. Not checked against a radio. The reset procedures could not be read and are left out. Yaesu and the model names here are trademarks of Yaesu. This card is an independent reference and is not published by, endorsed by or affiliated with Yaesu.

FT-857D

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT A YAESU PUBLICATION AND NOT AFFILIATED WITH YAESU.

MENU AND PANEL KEYS

WHAT THE BUTTONS DO

POWER	Hold one second to switch on or off. A tap engages fast tuning.
F or FUNC	Tap changes the multi-function row. Hold one second enters the menu.
A, B, C	Act on the displayed row. There are seventeen rows, lettered a to q.
SELECT knob	Detented. Tunes, picks memories, and scrolls the rows and menu items.
MAIN DIAL	Tuning, and setting menu values.
DSP	Tap jumps to the noise reduction row. Hold one second reaches the microphone equaliser.
MODE up / down	LSB, USB, CW, CW reverse, AM, FM, digital, packet.
BAND up / down	One band per tap.
V/M	Switches VFO and memory. Hold to store the VFO contents.
HOME	Recalls the home memory.
CLAR	Tap turns on the clarifier, plus or minus 9.99 kHz. Hold one second turns on IF shift.
LOCK	Locks the front panel.
SQL / RF	RF gain or squelch, set by menu item 080.

MULTI-FUNCTION ROWS

TAP F, THEN SELECT TO SCROLL

a	A/B, A=B, SPL VFO swap, equalise, and split.
b	MW, SKIP, TAG Memory write, scan skip, and channel naming.
d	RPT, REV, VOX Repeater shift, reverse, and VOX.
e	TON, ENC, TDCH CTCSS encoder and tone settings.
g	SCN, PRI, DW Scan, priority, and dual watch.
p	DNR, DNF, DBF Digital noise reduction, notch, and bandpass.

FILTERS

YF-122S

Optional Collins SSB filter, 2.3 kHz.

YF-122C

Optional Collins CW filter, 500 Hz.

YF-122CN

Optional Collins CW filter, 300 Hz.

SPECIFICATIONS

COVERAGE, RECEIVE	0.1 to 56 MHz, 76 to 108, 118 to 164, and 420 to 470 MHz
COVERAGE, TRANSMIT	160 through 6 m including 60 m on US sets, plus 2 m and 70 cm
MODES	CW, AM, SSB, FM, and packet at 1200 and 9600 baud
OUTPUT	100 W HF and 6 m, 50 W on 2 m, 20 W on 70 cm. AM carrier 25, 12.5 and 5 W
SUPPLY	13.8 V DC plus or minus 15 per cent, negative ground
CURRENT, TRANSMIT	22 A
CURRENT, RECEIVE	1 A, about 550 mA squelched
ANTENNA CONNECTORS	Two SO-239, one for HF and 6 m, one for 2 m and 70 cm
WEIGHT	About 4.6 lb
INTERNAL TUNER	None. The external FC-30 or an ATAS antenna is the option
COMPUTER CONTROL	CAT and linear jack, 8-pin mini DIN, shared with the tuner and amplifier

COMMON OPERATIONS

STORE A MEMORY

Tap F to the memory row, tap MW for memory check, then hold MW one second to write the VFO into the selected register.

RECALL A MEMORY

Tap V/M into memory mode, then SELECT to the channel.

SCAN

Tap F to the scan row, tap SCN to start scanning upward. Tap again to stop.

SET SPLIT

Tap F to the VFO row, tap SPL.

WORK A REPEATER

Tap F to the repeater row. RPT sets the shift direction; hold RPT one second to reach menu 076 for the shift size. Hold the tone key one second to reach menu 083 for the tone frequency.

RESET, MEMORIES KEPT

Hold the function key while switching the power on. Menu settings return to default and the memory channels survive. This is the one to reach for in the field.

RESET, MEMORIES ERASED

Hold V/M while switching the power on. Memories go, along with a handful of menu items.

RESET, EVERYTHING

Hold HOME while switching the power on. A full processor reset of memories and settings.

FT-891

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT A YAESU PUBLICATION AND NOT AFFILIATED WITH YAESU.

SPECIFICATIONS

COVERAGE, RECEIVE	0.03 to 56 MHz
COVERAGE, TRANSMIT	160 through 10 m including 60 m and the WARC bands, plus 6 m
MODES	SSB, CW, AM, FM
OUTPUT	100 W on SSB, CW and FM; 25 W AM
SUPPLY	13.8 V DC, plus or minus 15 percent
CURRENT, TRANSMIT	About 15 A, 23 A maximum
ANTENNA CONNECTOR	SO-239
WEIGHT	About 4.2 lb

OPERATING NOTES

- Fifteen amps key down is a field power problem, not a radio problem. Size the battery, the fuse and the wire for the current the radio actually draws, then run it at the power the contact needs.
- A hundred watts from a portable battery is a short outing. Most of what this radio does well in the field it does at thirty or forty.
- It has no internal tuner. Whatever you put up has to be resonant or matched by something you carried in.
- The detachable head is a connector, and connectors fail. Carry the spare cable or accept that a bad day ends early.

NOTES ON THIS RADIO

STATION SETTINGS

FILL THIS IN ONCE

CAT BAUD	
FILTERS AND PASSBAND	
CW PITCH AND SIDETONE	
KEYER SPEED	
MIC GAIN AND PROCESSOR	
SSB POWER, FIELD	
CW POWER, FIELD	
BREAK-IN SETTING	
EXTERNAL TUNER	
MEMORY BANK SCHEME	
RESET KEY SEQUENCE	
FIRMWARE VERSION	

MEMORY PLAN

WRITE IT DOWN BEFORE YOU NEED IT

[illegible]

Compiled from Yaesu's published specifications, September 2026. Not checked against a radio, and the operating manual was not read, so the panel is left for you to fill in. Yaesu and FT-891 are trademarks of Yaesu. This card is an independent reference and is not published by, endorsed by or affiliated with Yaesu.

FT-897D

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT A YAESU PUBLICATION AND NOT AFFILIATED WITH YAESU.

MENU AND PANEL KEYS

WHAT THE BUTTONS DO

POWER	Hold one second to switch on or off. A tap engages fast tuning.
F or FUNC	Tap changes the multi-function row. Hold one second enters the menu.
A, B, C	Act on the displayed row. There are seventeen rows, lettered a to q.
SELECT knob	Detented. Tunes, picks memories, and scrolls the rows and menu items.
MAIN DIAL	Tuning, and setting menu values.
DSP	Tap jumps to the noise reduction row. Hold one second reaches the microphone equaliser.
MODE up / down	LSB, USB, CW, CW reverse, AM, FM, digital, packet.
BAND up / down	One band per tap.
V/M	Switches VFO and memory. Hold to store the VFO contents.
HOME	Recalls the home memory.
CLAR	Tap turns on the clarifier, plus or minus 9.99 kHz. Hold one second turns on IF shift.
LOCK	Locks the front panel.
SQL / RF	RF gain or squelch, set by menu item 080.

MULTI-FUNCTION ROWS

TAP F, THEN SELECT TO SCROLL

a	A/B, A=B, SPL VFO swap, equalise, and split.
b	MW, SKIP, TAG Memory write, scan skip, and channel naming.
d	RPT, REV, VOX Repeater shift, reverse, and VOX.
e	TON, ENC, TDCH CTCSS encoder and tone settings.
g	SCN, PRI, DW Scan, priority, and dual watch.
p	DNR, DNF, DBF Digital noise reduction, notch, and bandpass.

FILTERS

YF-122S

Optional Collins SSB filter, 2.3 kHz.

YF-122C

Optional Collins CW filter, 500 Hz.

YF-122CN

Optional Collins CW filter, 300 Hz.

SPECIFICATIONS

COVERAGE, RECEIVE	0.1 to 56 MHz, 76 to 108, 118 to 164, and 420 to 470 MHz
COVERAGE, TRANSMIT	160 through 6 m including 60 m on US sets, plus 2 m and 70 cm
MODES	CW, AM, SSB, FM, and packet at 1200 and 9600 baud
OUTPUT	100 W HF and 6 m, 50 W on 2 m, 20 W on 70 cm. AM carrier 25, 12.5 and 5 W
OUTPUT, ON THE INTERNAL PACK	Drops automatically to 20 W, and 10 W on 70 cm
SUPPLY	13.8 V DC plus or minus 15 per cent. The optional internal pack is 13.2 V, 4500 mAh
CURRENT, TRANSMIT	22 A
CURRENT, RECEIVE	1 A, about 600 mA squelched
ANTENNA CONNECTORS	Two SO-239, one for HF and 6 m, one for 2 m and 70 cm
WEIGHT	About 8.6 lb without the battery, antenna or microphone
INTERNAL TUNER	None. A menu item chooses between an external tuner and an ATAS antenna
COMPUTER CONTROL	CAT and linear jack, 8-pin mini DIN, shared with the tuner and ATAS

COMMON OPERATIONS

STORE A MEMORY

Tap F to the memory row, tap MW for memory check, then hold MW one second to write the VFO into the selected register.

RECALL A MEMORY

Tap V/M into memory mode, then SELECT to the channel.

SCAN

Tap F to the scan row, tap SCN to start scanning upward. Tap again to stop.

SET SPLIT

Tap F to the VFO row, tap SPL.

WORK A REPEATER

Tap F to the repeater row. RPT sets the shift direction; hold RPT one second to reach menu 076 for the shift size. Hold the tone key one second to reach menu 083 for the tone frequency.

RESET, MEMORIES KEPT

Hold the function key while switching the power on. Menu settings return to default and the memory channels survive. This is the one to reach for in the field.

RESET, MEMORIES ERASED

Hold V/M while switching the power on. Memories go, along with a handful of menu items.

RESET, EVERYTHING

Hold HOME while switching the power on. A full processor reset of memories and settings.

FT-991A

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT A YAESU PUBLICATION AND NOT AFFILIATED WITH YAESU.

FILTERS**SSB**

Narrow default 1.5 kHz, adjustable from 200 Hz to 1.8 kHz.

CW

Narrow default 500 Hz, adjustable from 50 to 500 Hz.

RTTY AND DATA

Narrow default 300 Hz, adjustable from 50 to 500 Hz.

AM

Narrow default 2.4 kHz, adjustable from 1.8 to 3.0 kHz.

FM

500 Hz on the higher bands, with wider settings of 6, 9 and 16 kHz.

HOW TO SET IT

The narrow and wide touch key picks the mode-appropriate narrow filter. Touch the width key and rotate the multi knob to set the bandwidth.

SPECIFICATIONS

COVERAGE, RECEIVE	30 kHz to 56 MHz, 118 to 164 MHz, 420 to 470 MHz
COVERAGE, TRANSMIT	HF and 6 m, plus 2 m and 70 cm, amateur bands. Check your model's band edges
MODES	CW, AM, SSB, FM, RTTY, C4FM, and packet at 1200 and 9600 baud
OUTPUT	5 to 100 W on HF and 6 m, 5 to 50 W on 2 m and 70 cm. AM carrier 2 to 25 W
SUPPLY	13.8 V DC, negative ground
CURRENT, TRANSMIT	23 A at 100 W, 15 A at 50 W on VHF and UHF
CURRENT, RECEIVE	1.8 A with no signal, 2.2 A with one
WEIGHT	About 4.3 kg
INTERNAL TUNER	Yes, automatic, with a hundred channel memory
COMPUTER CONTROL	USB for CAT and firmware, plus a separate GPS and CAT serial connector

OPERATING NOTES

- Twenty-three amps key down. A hundred watts from a portable supply is a short outing whatever the radio.
- It has a real internal tuner, which is unusual in this class and worth knowing before you plan an antenna.
- One cable does CAT and audio, which makes it the easiest of these radios to get onto digital modes and the easiest to key twice by accident.
- The panel section of this card is empty on purpose. Yaesu's operating manual could not be read for it, and a guessed key sequence is worse than a blank line.

CAT AND COMPUTER CONTROL**USB PORT**

Straight to the computer. The radio carries a USB to dual serial bridge, so one cable gives both CAT and the sound card ports

DRIVER

Yaesu's USB driver has to be installed first

SERIAL CONNECTOR

The rear GPS and CAT connector, switched to serial by menu 028

MENU 031, CAT RATE

4800, 9600, 19200 or 38400. Yaesu does not publish a default. Write yours below

MENU 032, CAT TIMEOUT

10, 100, 1000 or 3000 milliseconds

MENU 033, CAT RTS

Enable or disable

STATION SETTINGS

FILL THIS IN ONCE

CAT RATE, MENU 031	
CAT TIMEOUT, MENU 032	
GPS OR SERIAL, MENU 028	
FILTER WIDTHS, PER MODE	
CW PITCH	
KEYER SPEED	
MIC GAIN	
SSB POWER, FIELD	
CW POWER, FIELD	
BREAK-IN SETTING	
REPEATER SHIFT AND TONE	
INTERNAL TUNER MEMORY	
RESET PROCEDURE, FROM THE MANUAL	
FIRMWARE VERSION	

FIELD REFERENCE, CONTINUED

NOTES ON THIS RADIO

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G90**FIELD REFERENCE**

INDEPENDENT REFERENCE. NOT A XIEGU PUBLICATION AND NOT AFFILIATED WITH XIEGU.

MENU AND PANEL KEYS**WHAT THE BUTTONS DO**

FUNC	Lights an F indicator, then press the target key for its second function. Press FUNC again to leave. Hold FUNC to open the system setup.
PRE / ATT	Cycles preamp, attenuator, and straight through.
CMP / F-L	Speech compression. With FUNC, the low-end cut of the digital filter.
NB / F-H	Noise blanker. With FUNC, the high-end cut of the digital filter.
AGC / SPL	Cycles AGC off, slow, fast and auto. With FUNC, split.
VM	Frequency mode or channel mode.
MW / MC	Channel store. With FUNC, channel clear.
A/B, A>B	Swaps the VFOs. With FUNC, copies the current VFO to the background one.
TUNE	Tap puts the tuner in line. Hold starts a tuning cycle.
Po	Transmit power. With FUNC, mic gain. See the note about the SWR sweep.
SWR THR	Standing wave protection threshold. With FUNC, audio input select.
LOCK	Backlight brightness, five levels. With FUNC, the spectrum reference level. Hold to lock the panel.
Main knob	Rotate to tune. Short press picks 100 Hz, 1 kHz or 10 kHz step.
Multifunction knob	Steps the 100 kHz digit by default. Hold to reassign it to squelch, power, keyer speed or display scale.

FILTERS**DIGITAL FILTER**

Shaped by the low cut and high cut, reached with FUNC over the compression and blanker keys. Xiegu prints no bandwidth figures anywhere in the manual.

SPECIFICATIONS

COVERAGE, RECEIVE	0.5 to 30 MHz
COVERAGE, TRANSMIT	160 through 10 m, amateur bands only
MODES	CW, AM, USB, LSB, FM
OUTPUT	20 W on SSB, CW and FM. 5 W AM carrier at 13.8 V. Set in 1 W steps
SUPPLY	10.5 to 16.5 V DC, negative ground
CURRENT, TRANSMIT	8 A maximum
CURRENT, RECEIVE	500 mA maximum
ANTENNA CONNECTOR	SL16-K, 50 ohms. This is the SO-239 family
WEIGHT	About 1.63 kg, case only
INTERNAL TUNER	Yes, built in, run from the TUNE key
COMPUTER CONTROL	COMM port, standard CI-V command set
STABILITY	Plus or minus 10 ppm after warm-up, 1 ppm per hour at 25 C

COMMON OPERATIONS**STORE A MEMORY**

Press MW, rotate the main knob to the channel number, short press the main knob to commit.

RECALL A MEMORY

Press VM to enter channel mode, then rotate the main knob.

CLEAR A MEMORY

FUNC then MW for clear mode, rotate to the channel, press the main knob.

SET SPLIT

Set the receive frequency and mode on VFO A, the transmit frequency on VFO B, then FUNC and the AGC key.

SET OUTPUT POWER

Short press Po, then rotate the multifunction knob. One watt per step.

RUN THE TUNER

Short press TUNE to bring it in line. Hold TUNE about a second to start tuning.

OPERATING NOTES

- Twenty watts from a radio this size is generous, and eight amps key down is the price. Fuse and wire for eight, not for what you usually run.
- The manual contradicts itself over the second function of the power key: one section gives mic gain, another gives the antenna sweep. Try it once at home and write down which one your radio does.
- There is no receive scan on this radio. The only sweep it has is a standing wave sweep of the antenna.
- Two entries in Xiegu's manual read 'to be determined'. The CW auto pager needs the GSOC controller and does nothing on a bare radio.
- Without the CE-19 expansion there is no band data output, so an amplifier will not follow the radio.
- Xiegu's manual describes no repeater offset, shift, tone or tone squelch for this radio. It has FM but no documented repeater support, so nothing about working a repeater appears on this card.

COMPUTER CONTROL**PORT**

COMM, using the standard CI-V command set

BAUD

Xiegu's manual states none. Find yours and write it below.

ACCESSORY PORT

ACC, 8-pin mini DIN, for computer audio in and out

OTHER JACKS

Headphone, key, and an I and Q output, all 3.5 mm

EXPANSION

The CE-19 adds PTT and band signal output, audio in and out, and data output on narrow FM

FIELD REFERENCE, CONTINUED

FILL THIS IN ONCE

WRITE IT DOWN BEFORE YOU NEED IT

CH	FREQUENCY	MODE	SHIFT / TONE	WHAT IT IS

X5105

FIELD REFERENCE

INDEPENDENT REFERENCE. NOT A XIEGU PUBLICATION AND NOT AFFILIATED WITH XIEGU.

MENU AND PANEL KEYS

WHAT THE BUTTONS DO

Power	Long press to switch on or off. Held more than eight seconds it resets the controller, which is the recovery for a crash.
MENU	Short press steps to the next menu page.
Soft keys	Four keys that operate whatever the screen shows above each one.
MODE	Cycles LSB, USB, CW, CW reverse, narrow FM and AM.
PRE / ATT	Cycles preamp on, attenuator on, and both off.
RIT	Receive incremental tuning.
NB	Toggles the noise blanker. Long press switches the battery display between level and voltage.
ATU	Short press brings the tuner in line, an antenna icon appears. Hold about a second to tune.
Po	Short press enters power setting, then the main knob, 0.1 to 5 W.
A/B	Swaps the VFOs. Long press copies the foreground VFO to the background.
V/M	Swaps VFO and memory. Long press while in memory writes the channel into the VFO.
LOCK	The manual gives two conflicting accounts. Either short press locks and a one second press toggles the backlight, or long press locks and short press steps the backlight. Find out which yours does.
UP / DN	Steps the band.

MENU PAGES

TAP MENU TO STEP THEM

1 COMMON	AGC off, slow, fast or auto. RF gain. Squelch. Compression with its level.
2 DSP	Digital filter with low and high cut. Noise reduction with level. Notch with centre and bandwidth. Equaliser, bass, mid and treble.
3 MEMORY	Channel manager, preset messages, and the boot screen callsign.
4 CW	Key type, sidetone, break-in time and break-in on or off. Iambic mode, keyer speed and dot to dash ratio in the submenu.
5 SPECIAL	Data mode with PSK31. The scanning receiver. The standing wave sweep.
6 INTERFACES	Mic selection and VOX. Speaker or earphone. Line gains. PTT mode.

FILTERS

DIGITAL FILTER

On or off, shaped by a low cut and a high cut. Xiegu prints no bandwidth figures.

NOTCH

Digital notch with a centre and a bandwidth.

NOISE REDUCTION

On or off, with a level.

EQUALISER

Receive bass, mid and treble.

SPECIFICATIONS

COVERAGE, RECEIVE	0.5 to 30 MHz and 50 to 54 MHz
COVERAGE, TRANSMIT	160 through 10 m, plus 6 m. Check the WARC bands on your own radio
MODES	CW, AM, USB, LSB, FM. The mode key cycles LSB, USB, CW, CW reverse, narrow FM and AM
OUTPUT	5 W on SSB, CW and FM. 1.5 W AM carrier. Adjustable from 0.1 W
SUPPLY	9 to 15 V DC, negative ground
CURRENT, TRANSMIT	2.5 A maximum
CURRENT, RECEIVE	600 mA maximum
ANTENNA CONNECTOR	BNC, 50 ohms
WEIGHT	About 0.94 kg, main unit only
INTERNAL TUNER	Yes, built in and automatic
INTERNAL BATTERY	3800 mAh at 12 V
COMPUTER CONTROL	COM, also marked CIV, using the standard CI-V command set

COMMON OPERATIONS

STORE A MEMORY

MENU to page 3, open the channel manager, select the channel number, save. Names are set the same way, stepping to the tag field and picking characters with the main knob.

RECALL A MEMORY

V/M switches into memory mode. A long press of V/M in memory copies the channel into the VFO.

SCAN

MENU to page 5, start the scanning receiver. Step is selectable at 0.5, 1, 2, 5 or 10 kHz, with pause and resume.

SET OUTPUT POWER

Short press Po, rotate the main knob, 0.1 to 5 W.

RUN THE TUNER

Short press ATU to bring it in line, hold about a second to tune. It returns to receive on its own.

SWEEP THE ANTENNA

MENU to page 5, standing wave sweep, with bandwidth, mode and speed options.

FIELD REFERENCE, CONTINUED

FILL THIS IN ONCE

- [illegible]

X6100**FIELD REFERENCE**

INDEPENDENT REFERENCE. NOT A XIEGU PUBLICATION AND NOT AFFILIATED WITH XIEGU.

MENU AND PANEL KEYS**WHAT THE BUTTONS DO**

GEN	Opens the general settings menu.
KEY	Opens the keyer settings.
DFN	Opens the digital function menu: noise reduction, blanker and notch.
APP	Opens the function menu.
MSG	Message editing and storage.
Soft keys	A row under the screen that does whatever the display shows above each one.
MFK	Multi-function knob. Long press about a second to unlock.
VOL / SQL / RFG	Rotate for volume. Press once for squelch depth, again for RF gain.
V/M	Switches VFO and memory.
AGC	AGC on and speed.
FST	Fast tuning step.
ATU	Tap brings the tuner in line. Hold about a second starts tuning.
PRE	Preamp and attenuator.
A/B	Swaps the VFOs.
Lock	Long press about a second locks the panel.

MENU MAP**THE LAUNCHER KEYS OPEN THESE**

GEN 1 RADIO SETTING 1	AGC knee, slope and hang. TX POWER. Mic selection and gain. Line levels. Monitor level. PTT mode. Band stack. Charger on or off.
GEN 2 RADIO SETTING 2	RIT, XIT, split, and the IQ output.
GEN 3 DISPLAY SETTING	Spectrum and waterfall averaging and reference, and backlight level.
GEN 4 SYSTEM SETTING	Time, and firmware upgrade.
GEN 5 MEMORY SETTING	Save the VFO, tag it, mark it, erase it.
KEY KEYER	Key type, speed, iambic mode, sidetone frequency and level, break-in time, dot to dash ratio, and a CW trainer.
DFN DIGITAL	Noise reduction and depth, blanker with width and level, and a digital notch with centre and width.

FILTERS**FILTER1 TO FILTER3**

Three selectable filters, shaped by a high cut and a low cut, with a default that restores them. Xiegu prints no bandwidth figures.

SPECIFICATIONS

COVERAGE, RECEIVE	0.5 to 30 MHz and 50 to 54 MHz
COVERAGE, TRANSMIT	160 through 10 m, plus 6 m
MODES	CW, AM, SSB, FM
OUTPUT, EXTERNAL DC	10 W on SSB, CW and FM. 2.5 W AM carrier
OUTPUT, INTERNAL BATTERY	5 W on SSB, CW and FM. 1.5 W AM carrier. The supply halves it
SUPPLY	9 to 15 V DC, negative ground
CURRENT, TRANSMIT	3 A maximum
CURRENT, RECEIVE	330 mA maximum
ANTENNA CONNECTOR	BNC, 50 ohms
WEIGHT	About 880 g, host only
INTERNAL TUNER	Yes. Matches to about 4.5 to 1, first tune within 15 s
INTERNAL BATTERY	3000 mAh at 12 V. About six hours to charge
COMPUTER CONTROL	USB Type-C, with the port driver installed

COMMON OPERATIONS**STORE A MEMORY**

General settings, memory setting, save VFO. Name it with the tag item.

RECALL A MEMORY

V/M switches VFO and memory. The left and right rockers step channels.

ERASE A MEMORY

General settings, memory setting, erase memo.

SET SPLIT

General settings, radio setting 2, split.

SET OUTPUT POWER

General settings, radio setting 1, TX power.

RUN THE TUNER

Short press ATU to bring it in line, long press about a second to tune.

FACTORY RESET

Rotate the multi-function knob to factory reset, press it, confirm. The radio shuts down and needs turning on again. Xiegu does not say whether memory channels survive, so photograph them first.

OPERATING NOTES

- Ten watts needs external power. On the internal pack it makes five, and the radio does not announce the change.
- There is no receive scan on this radio. The only sweep it has is a standing wave sweep.
- The control software wants the radio selected as an X108G, which is not obvious and costs people an evening.
- Everything on this card is stock firmware. Third-party firmware changes the keys, the menus and the labels, and none of it is in Xiegu's documentation.
- Xiegu's manual describes no repeater offset, shift, tone or tone squelch for this radio. It has FM but no documented repeater support, so nothing about working a repeater appears on this card.

FIELD REFERENCE, CONTINUED

WRITE IT DOWN BEFORE YOU NEED IT

CH	FREQUENCY	MODE	SHIFT / TONE	WHAT IT IS
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FILL THIS IN ONCE

COLOPHON

The interior is set in IBM Plex: Plex Serif for reading matter, Plex Sans and Plex Sans Condensed for labels and rules, Plex Mono for tabular matter. Log grids are ruled at 0.4 point, a weight that carries pencil, ballpoint and fine-line ink without competing with the entry. Row heights are set for handwriting done in the field rather than at a desk.

Section markers are printed rather than die cut, so the book can be produced by any printer without specialist finishing.

Sources for Part One

- 47 CFR Part 97, Amateur Radio Service, US Federal Communications Commission.
- ARRL frequency allocations and band plan charts, and ARRL/RAC contest section list.
- FCC Report and Order on 60 meters, new 5351.5 to 5366.5 kHz subband effective 13 February 2026.
- FCC Report and Order eliminating HF symbol rate limits in favor of a 2.8 kHz bandwidth limit, effective 2024.
- Parks on the Air program rules and activator documentation.
- Summits on the Air general rules.
- FCC Office of Engineering and Technology Bulletin 65 on RF exposure, and the amateur exposure evaluation requirement in effect since 3 May 2023.

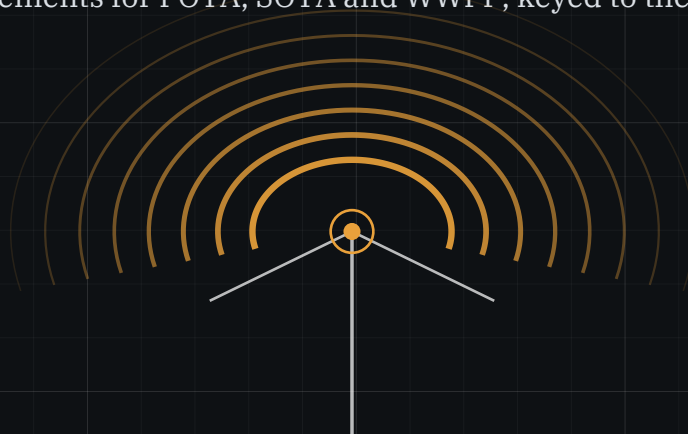
Coax loss, velocity factor, and conductor ampacity figures are nominal values for typical cable and wire of each type at room temperature. Use them for planning. Measure for anything that matters.

A logbook for a station that is carried in, assembled on site, and taken down the same day.

FIELD RECORD is organized around the outing rather than the contact. Twenty-six deployments, four pages each: a Site Record completed before the first call, two sheets of contacts in the columns portable operation requires, and a debrief written at the site. Part One is a field reference compiled from primary sources for operators working away from mains power. Part Three holds the station records that carry forward from one volume to the next.

IN PART ONE

- US band plan and privileges in color, including the 60 meter allocation as revised in February 2026
- South Carolina repeaters and nets, and a quick reference for the Icom IC-706MKII
- Antenna field math, coax loss, SWR tables, and power budget worksheets
- RF exposure limits and the station evaluation required of every licensee
- Propagation indices, grid squares, Q signals, prosigns, phonetics, and ARRL sections
- Activation requirements for POTA, SOTA and WWFF, keyed to the UTC day



SAMPLE "NOCALL" STATION

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